

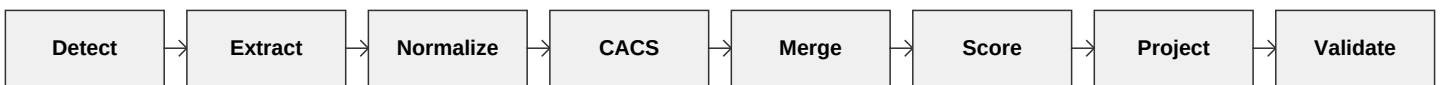
System Architecture & Design Document

Multi-Source Candidate Data Transformer Pipeline

1. Problem Statement & Objective

The candidate data ingestion pipeline aggregates records from disparate structured (ATS, CSV, GitHub) and unstructured (Resumes, Notes) origins. The primary objective is to design a deterministic ETL (Extract, Transform, Load) pipeline that standardizes schema variability, resolves conflicting data points across overlapping sources, and emits a single canonical profile. Downstream consumers require high data fidelity, explicit provenance tracking, and configurable payload projections.

2. System Architecture



The pipeline operates as a stateless, multi-stage DAG to ensure idempotency and determinism:

1. Ingestion & Extraction: Parsers ingest payloads and map them to a unified internal representation (CandidateFragment).
2. Normalization: Applies global formatting standards (E.164 for telephony, ISO-3166 for locales, canonical mappings for tech stacks).
3. Conflict Analysis (CACS): A pre-merge analytical heuristic that detects data divergence, applies temporal recency stratification to skills, and generates an auditable Lineage IR.
4. Merge & Deduplication: Resolves identity collisions via primary key matching. Array fields are deduplicated using Levenshtein distance metrics (rapidfuzz).
5. Confidence Scoring: Aggregates source-level reliability weights and cross-validation bonuses to assign a confidence index [0.0 - 1.0].
6. Projection: Hydrates the final canonical payload according to dynamic downstream JSON schemas.

3. Core Resolution Heuristics

- Identity Resolution: Evaluates overlapping candidate entities using primary keys (email) and secondary fallback permutations (normalized name + phone).
- Array Deduplication: Employs fuzzy string matching (threshold > 75%) coupled with date range intersections to merge overlapping experience and education records.
- Unstructured Contextual Parsing: Utilizes strict regex boundaries and contextual anchors rather than stochastic LLMs, drastically reducing hallucination risk when parsing free-text roles and institutions.

4. Interfaces & Extensibility

- Command Line Interface (CLI): Engineered for batch processing, CI/CD integration, and high-throughput execution.
- Web UI (Streamlit): Provides an interactive dashboard for QA and product evaluators to visually validate

System Architecture & Design Document

Multi-Source Candidate Data Transformer Pipeline

merging heuristics and projected schemas in real-time.